Type 2: Application Project

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Smart Home HAR for Speech-Impaired & Hearing-Impaired Users

23CSE399 – Project Phase 1 | Panel Review 2

S.No	Register Number	Student Name
1	CB.SC.U4CSE23101	Adhikkesh S K
2	CB.SC.U4CSE23102	Adithya Ajay
3	CB.SC.U4CSE23105	Amrith B
4	CB.SC.U4CSE23119	Gowreesh B
5	CB.SC.U4CSE23236	Rishiikesh S K

Guide: **Ms. Nalinadevi K** Dept. of CSE, Amrita Vishwa Vidyapeetham

March 30, 2026

1a. Use Cases: Differentiating Action and Sign

How Multi-Modality Resolves Conflicting Sensor Interpretations

UC1: Kitchen (Action → Sign)

- **Scenario:** Cooking, then stops to sign to someone entering.
- **Ambient:** Sees ongoing presence.
- **Wearable:** Confuses sign with cooking arm motions.
- **WiFi:** Captures precise sign gesture.
- **Result:** Fusion identifies shift to **Communication**.

UC2: Video Call (Idle → Sign)

- **Scenario:** Sitting idle, gets a video call, begins signing.
- **Ambient:** Detects purely static/idle state.
- **Wearable:** Triggers on sudden, structured arm movement.
- **WiFi:** Translates movement to text.
- **Result:** Fusion confirms **Intent to Communicate**.

1b. Use Cases: Identifying True Idle State

Transitioning from Activity to Rest without False Positives

UC3: Resting (Action → Idle)

- **Scenario:** Finishes chores and sits down to rest.
- **Ambient:** Motion sensors deactivate as room settles.
- **Wearable:** Macro-acceleration drops to zero.
- **WiFi:** Gates OFF to save compute power.
- **Result:** Fusion safely logs user state as completely **Idle**.

Key Insights: The Necessity of Fusion

- **Transition Problem:** Single sensors misclassify **Action** → **Sign** as generic movement.
- **Isolating Intent:** Fusion clearly separates **Communication** from daily chores.
- **System Efficiency:** Prevents false triggers, letting the system safely rest.

2. Motivation: Why This Project Matters

Blending Real-World Needs with Smart Sensor Innovation

1. Beyond Voice Commands

- Smart hubs **exclude non-verbal users**.
- We focus on **understanding silent intent**.

2. No Cameras or Suits

- Cameras severely **invade home privacy**.
- Daily sensor suits are **highly obtrusive**.

3. Fixing Sensor Confusion

- Single sensors **confuse chores and signs**.
- We cleanly **separate distinct actions**.

4. A New Use for WiFi

- We repurpose WiFi for **sign language**.
- Enables **camera-free gesture tracking**.

5. Smart Sensor Fusion

- Fusing ambient and WiFi **predicts intent**.
- System adapts to **complex user behavior**.

Key Insight: Our motivation is to build an **un,obstrusive, multi-modal system** that safely translates **silent intent** without disrupting daily life.

3. Problem Statement & Research Objectives

Addressing the Gap in Non-Verbal Smart Home Actuation

Problem Statement: Current smart home systems either compromise privacy through camera usage or rely on single-modality sensors, limiting their ability to distinguish between activities, sign language gestures, and idle states, and to recognize parallel and group activities in real-world scenarios. Our motivation is to build an unobstrusive, multi-modal system that safely translates silent intent without disrupting daily life.

Translation to Sub-Problems (Research Objectives):

- **1. Dataset Preparation:** Preparation of a dataset capturing **simultaneous and multi-user activities**.
- **2. Model Fusion:** **Fusion** of models to capture both **spatial and temporal patterns** in the data.
- **3. State Identification:** Differentiating **Activity, Sign, and Idle States**.
- **4. Silent Communication:** **Sign Language to Text** Translation.

4. Dataset Survey: What Data Exists and What It Lacks

Why a Custom Multi-Modal Dataset Is Needed

Type	Dataset	What It Captures	Key Gap
Ambient	CASAS	Real apartment PIR & door sensors	Cannot identify specific users
Ambient	ARAS	Multi-resident ambient sensor logs	Lacks per-person ground truth
Wearable	UCI-HAR	Smartphone acceleration for ADLs	Blind to indoor location/context
Wearable	Opportunity	72 on-body and object sensors	Highly impractical for daily living
WiFi	Widar 3.0	Large-scale WiFi CSI gestures	Fails with simultaneous user motion
WiFi	SignFi	WiFi CSI for ASL gestures	Breaks during concurrent body movement

Takeaway: No existing dataset simultaneously fuses ambient, wearable, and WiFi CSI. This critical gap forces a trade-off between privacy and granularity, **directly necessitating our custom data collection in Phase 2**.

5. Product Survey: Commercial Systems & Their Gaps

Why No Existing Product Solves the Accessibility Problem

Product	What It Does	Why It Fails Speech / Hearing-Impaired Users
Smart Hubs (Nest, Alexa)	Voice commands & simple automation	Excludes speech-impaired users; no sign language support
Wearables (Fitbit)	Heart rate, steps, outdoor GPS	No indoor context; cannot interact with the home environment
Security Systems (ADT, Ring)	Perimeter cameras & door sensors	Invasive cameras in personal spaces; binary non-camera sensors
Elderly Monitors (LifeAlert)	Emergency button press after a fall	Purely reactive — triggers after accident; cannot predict incidents
WiFi Sensing (Origin Wireless)	Device-free presence via WiFi	Cannot distinguish multiple people; mixed signals cause confusion

Takeaway: Every commercial product requires voice, uses cameras, or ignores sign language. None offer a privacy-safe, silent communication channel for speech- or hearing-impaired residents.

6. Research Survey: State of the Art & Open Problems

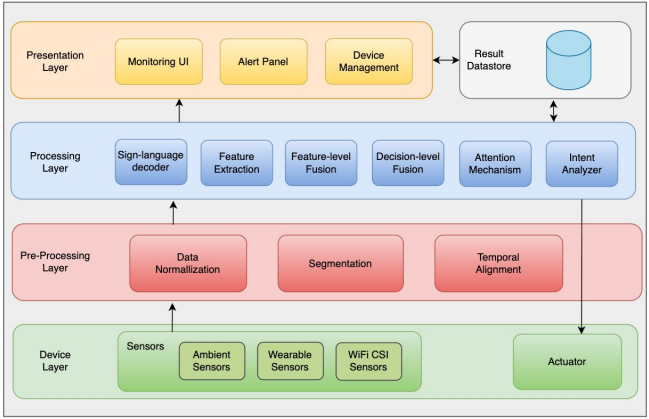
Key Papers and What They Leave Unsolved

Paper (Year)	Method / Technique	Description
HAR in Smart Living (2023)	ML/DL-based HAR using multi-sensor data	Comprehensive review of HAR techniques for smart living applications.
IoT Smart Home Automation (2021)	IoT-based automation using sensors and APIs	Proposes an IoT system for automating and controlling smart home devices.
Activity Predictors via Sensor Data (2017)	Imitation Learning with Regression models	Develops models to predict future human activities using sensor data.
SignFi: Sign Language via WiFi (2018)	CNN (9-layer) using WiFi CSI signals	Uses deep learning on WiFi signals to recognize sign language gestures.

Takeaway: Existing research treats IoT, predictive modeling, and WiFi sensing as isolated domains. We bridge this gap by unifying them into a single, multi-modal intent-aware framework.

7. System Architecture: Layered Architecture

From Raw Sensor Data to Activity Label and Smart Home Actuation



A four-tier layered system architecture

Key Insight: Synchronized ambient, wearable, and WiFi streams are fused to trigger precise smart home

8. Implementation 1: Ambient Modality

Evaluating Ambient Modality to Gate Multi-Modal Transitions

- **Why Ambient Sensors:** Tracks basic room activity safely without using cameras.
- **Why CASAS Data?:** Gives real home sensor logs (like motion and doors) to match daily life.
- **Implementation & Graph:** The graph compares models across datasets, showing Bi-LSTM and Ens2-LSTM generally achieve the highest accuracy.

LSTM Performance on CASAS

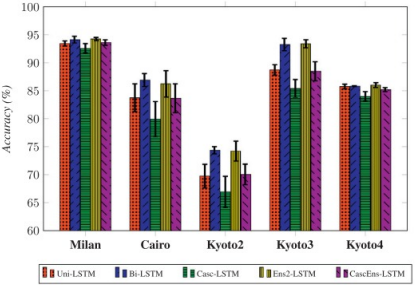


Fig. 9. Averaged accuracy and standard deviation of each dataset over all three folds.

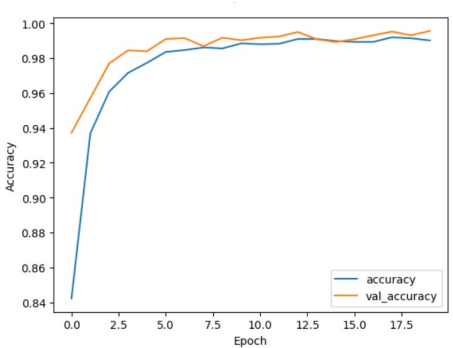
View Colab Notebook Implementation

8. Implementation 2: Wearable Modality

Type 2: Application Project

Evaluating IMU Modality for Spatial Activity Recognition

- **Significance of Wearables:** Captures fine-grained spatial micro-movements (e.g., posture changes, stirring) that ambient room sensors miss.
- **Why OPPORTUNITY Data? :** Supplies complex, raw multi-axis IMU streams covering both everyday micro-gestures and macro-sequences.
- **Implementation & Graphs:** Trained a CNN to extract physical patterns; the graphs show it reliably achieves **99.4% classification accuracy**.



View Colab Notebook Implementation

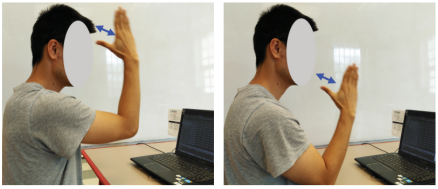
8. Study of WiFi CSI for Sign Language

Evaluating WiFi Modality for Non-Intrusive Sign Recognition

Type 2: Application Project

- Captures **wireless reflections** to eliminate **privacy-invasive cameras** and the **obtrusive style** of wearable sensors.
- Recovers **amplitude and phase** from raw CSI to extract **fine-grained gesture characteristics**.
- Uses a **9-layer CNN** to classify 276 distinct ASL gestures with **94% accuracy**.

Same handshape, different hand locations.



(a) Sign gesture for "Father" (b) Sign gesture for "Mother"

9. Future Implementation

Next Phase Objectives & System Expansion

Next Phase Objectives :

- **Proprietary Data Capture:** Construct a custom dataset fusing real-world ambient, wearable, and WiFi CSI streams.
- **Hybrid Fusion Models:** Develop an attention-guided architecture merging LSTM temporal and CNN spatial features.
- **Sign Language Recognition:** Integrate WiFi and wearable gesture tracking for users with speech/hearing disabilities.
- **End-User App Development:** Create a UI for real-time system monitoring, smart home interaction, and anomaly alerts.

References I

[1] D. J. Cook et al. *CASAS: A Smart Home in a Box*. Computer, 46(7), 2013. https://doi.org/10.1109/MC.2012.328

[2] H. Alemdar et al. *ARAS human activity datasets in multiple homes*. PervasiveHealth, 2013. https://ieeexplore.ieee.org/document/6563930

[3] D. Anguita et al. *A Public Domain Dataset for HAR Using Smartphones*. ESANN, 2013. https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones

[4] R. Chavarriaga et al. *The Opportunity challenge*. Pattern Recognition Letters, 34(15), 2013. https://doi.org/10.1016/j.patrec.2012.12.014

[5] D. Micucci et al. *UniMiB SHAR: A Dataset for HAR*. Applied Sciences, 7(10), 2017. https://doi.org/10.3390/app7101101

[6] Y. Zheng et al. *Widar3.0: Zero-Effort Cross-Domain Gesture Recognition*. IEEE TPAMI, 43(9), 2021. https://tns.thss.tsinghua.edu.cn/widar3.0/data/TPAMI_Widar3.0_paper.pdf

References II

[7] G. Zhu et al. *CSI-Bench: A Dataset for Multi-task WiFi Sensing*. arXiv:2505.21866, 2025.
`https://arxiv.org/abs/2505.21866`

[8] S. Huang et al. *WiMANS: A Benchmark Dataset for Multi-User WiFi Sensing*. arXiv:2402.09430, 2024.
`https://arxiv.org/abs/2402.09430`

[9] Y. Ma et al. *SignFi: Sign Language Recognition Using WiFi*. ACM IMWUT, 2(1), 2018.
`https://dl.acm.org/doi/10.1145/3191755`

[10] Wang et al. *TedNet: Device-free Real-time Fall Detection via Wi-Fi*. arXiv:2504.16655, 2025.
`https://arxiv.org/abs/2504.16655`

[11] J. Li et al. *WiFinger: Finger Gesture Recognition Using CSI*. UbiComp, 2016.
`https://dl.acm.org/doi/10.1145/2971648.2971690`

[12] B. D. Minor et al. *Learning Activity Predictors from Sensor Data*. IEEE TKDE, 29(12), 2017.
`https://ieeexplore.ieee.org/document/8031060`

Thank You

Smart Home HAR for Speech-Impaired & Hearing-Impaired Users

Dept. of CSE, Amrita Vishwa Vidyapeetham